- **The ASHRAE Great Energy Predictor III competition: Overview and results**

Link - **The ASHRAE Great Energy Predictor III competition: Overview and results: Science and Technology for the Built Environment: Vol 26, No 10**

Markdown-
The study is based on a large-scale machine learning competition where multiple approaches were tested for building energy prediction. The most successful methods used gradient boosting techniques, particularly LightGBM, often combined with ensemble approaches. These models were applied with extensive feature engineering and data preprocessing techniques.

Dataset-
The study uses the ASHRAE GEPIII dataset, which contains building energy consumption data from 1,448 buildings across 16 sites. The dataset includes over 20 million training samples and more than 61 million hourly energy readings collected between 2016 and 2018. It also includes building metadata and weather data.

Metric-
The performance of the models is evaluated using Root Mean Squared Logarithmic Error (RMSLE), which measures the difference between predicted and actual energy consumption values on a logarithmic scale.

Markdown-
The results show that gradient boosting models, especially LightGBM, achieved the best performance compared to other machine learning and deep learning approaches in the competition.

Strength-
The study uses a very large real-world dataset and demonstrates high model performance using efficient machine learning techniques such as LightGBM.

Weakness-
The study focuses mainly on tabular data, so the results may not generalise well to time-series or unstructured data where deep learning models may perform better.

Why LightGBM over Deep Learning-
LightGBM is preferred over deep learning in this study because it performs better on structured/tabular data, requires less computational cost, and is easier to train compared to deep learning models.

Dataset size:
Not explicitly mentioned; large-scale dataset with multiple buildings, time-series meter readings, weather and metadata features
Overlap:
Uses Gradient Boosting models (LightGBM) with extensive feature engineering
Relevance:
Demonstrates that LightGBM performs strongly on large real-world energy datasets

- ✓ **The most commonly used model was the LightGBM framework that is designed to be fast in training speed and low in memory usage, while still maintaining high accuracy.9 Other gradient boosting models used include CatBoost,10 XGBoost,11 and LiteMORT.12 A few other models were used in the large winning ensembles such as Multi-Layer Perceptrons, Feed□forward Neural Networks (FFNN), and Random Forest models. (As mentioned in the paper /1443)**

- ❖ **Gradient boosting machines and careful pre-processing work best: ASHRAE Great Energy Predictor III lessons learned**

Link - **Gradient boosting machines and careful pre-processing work best: ASHRAE Great Energy Predictor III lessons learned**

Method:

The study analyses top-performing solutions from the ASHRAE GEPIII competition using survey data and publicly shared notebooks. The results show that most high-performing models used ensemble approaches based on Gradient Boosting Machines (GBM), particularly LightGBM, combined with strong preprocessing and feature engineering techniques.

Dataset:

The study is based on the ASHRAE GEPIII competition dataset, which consists of large-scale building energy consumption data collected from multiple buildings and energy meters. The dataset includes historical energy readings, weather data, and building metadata used for long-term energy prediction tasks.

Metric:

The metric is not explicitly given in this paper; it is taken from the original ASHRAE competition, where RMSLE was used.(**Paper - The ASHRAE Great Energy Predictor III competition: Overview and results)**
Link - **The ASHRAE Great Energy Predictor III competition: Overview and results: Science and Technology for the Built Environment: Vol 26, No 10**

Key Finding:

The study finds that Gradient Boosting models, especially LightGBM, were the most effective approaches for building energy prediction, outperforming other machine learning and deep learning models.

Strength:

The study provides real-world insights from top-performing solutions and highlights practical best practices such as feature engineering and preprocessing for improving model performance.

Weakness:

The study is based on a specific competition dataset and may not generalise to all building energy prediction problems or other types of data.

Dataset size:
Based on ASHRAE building energy dataset; exact size not explicitly stated
Overlap:
Uses Gradient Boosting Machines (LightGBM) and ensemble learning
Relevance:
Shows boosting models outperform many traditional approaches in energy prediction

Justification:

LightGBM is preferred over deep learning models because it performs better on structured/tabular data, requires less computational cost, and achieves high accuracy with proper preprocessing and feature engineering.

- **Potential energy saving estimation for retrofit building with ASHRAE-Great Energy Predictor III using machine learning**

Link - **Microsoft Word - FQRBI2IHEN**

Method:

The study applies machine learning models including Linear Regression, Random Forest, and Gradient Boosting Machine (LightGBM). The problem is formulated as a regression task to predict building energy consumption. Feature engineering and preprocessing are performed on weather and building data before training the models.

Dataset:

The dataset used is the ASHRAE Great Energy Predictor III dataset obtained from Kaggle. It includes building energy consumption data along with weather data and building metadata such as site ID and timestamp. [ASHRAE Kaggle dataset (building + weather data)]

Metric:

The model performance is evaluated using Root Mean Squared Logarithmic Error (RMSLE), which is suitable for measuring prediction accuracy in energy consumption tasks.

Key Findings:

The study finds that ensemble models, especially LightGBM, outperform traditional models like Linear Regression in predicting building energy consumption. LightGBM achieves better accuracy due to its ability to capture complex nonlinear relationships.

Strength:

A key strength of the study is the use of advanced ensemble models like LightGBM, which improve prediction accuracy. The study also uses a large real-world dataset, making the results more reliable and practical.

Weakness:

One limitation of the study is that it relies heavily on data quality and preprocessing. Additionally, the model may require high computational resources, and results may not generalize well to different datasets without retraining.

Dataset size:
Uses ASHRAE dataset (large real-world building energy dataset; size not explicitly mentioned)

Overlap:
Uses LightGBM and feature engineering on building and weather data

Relevance:
Supports use of boosting models for real-world building energy prediction tasks

- **Prediction of heating and cooling loads based on light gradient boosting machine algorithms**

Link: **Prediction of heating and cooling loads based on light gradient boosting machine algorithms - ScienceDirect**

Method:

The study uses LightGBM with hyperparameter optimisation techniques such as Random Search, Grid Search, CMA-ES, and TPE. Feature selection is also applied**.**

Dataset:

The dataset consists of simulated residential building data including structural and environmental parameters for predicting heating and cooling loads.

Metric:

Model performance is evaluated using RMSE, MAE, and R^2.

Key Finding:

Optimised LightGBM significantly improves prediction accuracy compared to default models.

Strength:

Use of multiple optimisation techniques improves model accuracy and robustness.

Weakness:

Relies on simulated data and requires high computational resources.

Dataset size:
Uses building energy dataset (simulated/experimental); exact size not explicitly mentioned

Overlap:
Applies LightGBM with feature selection and optimisation

Relevance:
Shows LightGBM improves prediction accuracy for heating and cooling loads

- **Improving energy management practices through accurate building energy consumption prediction: analyzing the performance of LightGBM, RF, and XGBoost models with advanced optimization strategies**

  Link - Improving energy management practices through accurate building energy consumption prediction: analyzing the performance of LightGBM, RF, and XGBoost models with advanced optimization strategies | Electrical Engineering | Springer Nature Link

Method:

The study uses machine learning models including LightGBM, Random Forest (RF), and XGBoost. These models are further enhanced using optimization techniques such as Satin Bowerbird Optimizer (SBO) and Moth Flame Optimization (MFO).

Dataset:

The dataset is collected from two buildings located in Richland, Washington (USA), covering data from 2009 to 2011. It includes time-related features, weather data, and historical energy consumption variables.

Metric:

The model performance is evaluated using multiple metrics such as MAE, RMSE, MSE, and R^2.

Key Finding:

The hybrid LightGBM-SBO model achieved the best performance compared to other models, with higher accuracy and lower error.

Strength:

The study improves prediction accuracy by combining machine learning models with advanced optimization techniques. It also uses real-world building data.

Weakness:

The dataset is limited to specific buildings and time period, which may reduce generalizability. Also, optimization methods increase computational complexity.

Dataset size:
Uses historical building energy data collected over multiple years (e.g., 2009–2011); exact size not specified

Overlap:
Uses LightGBM and ensemble models

Relevance:
Compares models and highlights strong performance of LightGBM

- **The Comparison of Machine-Learning Methods XGBoost and LightGBM to Predict Energy Development**

  Link : The Comparison of Machine-Learning Methods XGBoost and LightGBM to Predict Energy Development | Springer Nature Link

Method:

The study uses two machine learning models: XGBoost and LightGBM for energy prediction.

Dataset:

The dataset consists of time series data collected from a thermal plant for energy prediction.

Metric:

The study evaluates model performance by comparing prediction accuracy between XGBoost and LightGBM.

Key Finding:

The study compares LightGBM and XGBoost and shows differences in prediction performance, highlighting the importance of feature engineering and model selection.

Strength:

The study directly compares two advanced machine learning models and applies feature engineering to improve prediction performance.

Weakness:

The study is limited to a specific dataset (thermal plant data), which may not generalize to other energy systems.

Dataset size:
Time-series building energy dataset; exact size not explicitly mentioned

Overlap:
Uses LightGBM and Gradient Boosting models

Relevance:
Shows LightGBM is efficient and competitive with other boosting models

- **Building energy consumption prediction for residential buildings using deep learning and other machine learning techniques**

  Link - Building energy consumption prediction for residential buildings using deep learning and other machine learning techniques - ScienceDirect

  Abstract

  The high proportion of energy consumed in buildings has engendered the manifestation of many environmental problems which deploy adverse impacts on the existence of mankind. The prediction of building energy use is essentially proclaimed to be a method for energy conservation and improved decision-making towards decreasing energy usage. Also, the construction of energy efficient buildings will aid the reduction of total energy consumed in newly constructed buildings. Machine Learning (ML) method is recognised as the best suited approach for producing desired outcomes in prediction task. Hence, in several studies, ML has been applied in the field of energy consumption of operational building. However, there are not many studies investigating the suitability of ML methods for forecasting the potential building energy consumption at the early design phase to reduce the construction of more energy inefficient buildings. To address this gap, this paper presents the utilization of several machine learning techniques namely Artificial Neural Network (ANN), Gradient Boosting (GB), Deep Neural Network (DNN), Random Forest (RF), Stacking, K Nearest Neighbour (KNN), Support Vector Machine (SVM), Decision tree (DT) and Linear Regression (LR) for predicting annual building energy consumption using a large dataset of residential buildings. This study also examines the effect of the building clusters on the model performance. The novelty of this paper is to develop a model that enables designers input key features of a building design and forecast the annual average energy consumption at the early stages of development. This result reveals DNN as the most efficient predictive model for energy use at the early design phase and this presents a motivation for building designers to utilize it before construction to make informed decision, manage and optimize design.

  Dataset details from the paper

- Source: Ministry of Housing Communities and Local Government (MHCLG) repository
- Building dataset: Metadata and energy consumption data for 5000 residential buildings in the UK
- Geographic scope: 10 postcode areas in the UK
- Distribution: 500 buildings from each of the 10 postcode areas
- Target/output variable: Annual energy consumption for the year 2020
- Meteorological data source: Meteostat
- Weather features: temperature, wind speed, pressure
- Weather granularity: daily data from 1 Jan 2020 to 31 Dec 2020, later averaged monthly
- After merging: total of 285,000 data points
- Missing building instances removed: 540 instances

- Smaller experiment mentioned later: after processing, 1000 instances became 908 due to deletion of 92 missing rows

Metric

The performance of each model is evaluated using the following performance measures: R-Squared (R2), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE).

Conclusion

This paper compared the prediction performance of nine machine learning techniques namely DNN, ANN, SVR, DT, GB, LR, Stacking, RF and KNN for predicting annual energy consumption. The good performance result generated in the study have proffered a model that enables building designer predict energy consumption at the early design phase. In general, DNN produced better results than other model for energy use prediction. However, there are other proven efficient predictive models in this study such as ANN, GB and SVM. In terms of computational efficiency, DT produced the best result with a training time of 1.2 s.

Subsequently, a sensitivity analysis was conducted to satisfy the third objective. Using the most dominant building cluster in the dataset, feature selection was implemented, and nine ML models were developed and evaluated. The result shows that the performance of the model is not sensitive to a specific type of building. Hence, building cluster has no significant effect on the feature selected and the performance result. Furthermore, nine ML models were developed using two different sizes of data to satisfy the fourth objective. The result shows that the size of data has significant effect on the performance of the model.

The high performance of these models presents a motivation for building designers to use it at the early design phase to make informed decision, manage and optimize design. Future research should focus on the applying DNN and other deep learning models on an even larger data of greater than 40,000 in anticipation of better performance. Also explore the suitability of other ensemble algorithms for predicting building energy consumption.

Dataset size:
Uses building energy dataset with multiple buildings and time-series data; size not explicitly mentioned

Overlap:
Includes Gradient Boosting models along with deep learning models

Relevance:
Shows ML models like boosting can perform as well as or better than deep learning

❖ **Tuning machine learning models for prediction of building energy loads**

Link – Tuning machine learning models for prediction of building energy loads - ScienceDirect

Abstract

It is of great significance to achieve the prediction of building energy consumption. However, machine learning, as a promising technique for many practical applications, was rarely utilized in this field. The most important reason is that the predictive structure with best performance is difficult to be determined. To fill the gap, this paper offers one in-depth review, which focuses on

the accuracy analyses and model comparisons. Specifically, the accuracy analyses were conducted based on different types of buildings (e.g. residential building, commercial building, government building or educational building), different type of temporal granularity (e.g. sub-hourly, hourly, daily or annual), as well as input/output variables and historical data collections. Further, artificial neural network (ANN) and support vector machine (SVM), as the epidemic models, were compared in terms of their complexity of prediction processes, accuracies of results, the amounts of required historical data, the numbers of inputs, etc. Then the hybrid and single machine learning methods were outlined and compared in terms of their strengths and weaknesses. In addition, several vital defects and further research directions are presented from a multivariate perspective. We hope that machine learning method could capture more attention from investigators via our introduction and perspective, due to its potential development of accuracy and reliability.

Dataset details from the paper

Two simulated building datasets were used: (1) an Ecotect dataset with 768 residential building variations and 8 envelope features, and (2) an EnergyPlus dataset containing residential and commercial buildings with 28 envelope and climate features. The prediction targets were heating and cooling loads.

Metric

The paper discusses several evaluation metrics such as CV, MBE, MSE, RMSE, MSPE, MAPE, and MAE, but the main metrics used in this study are RMSE, MAE, and R^2.

Conclusion

The research presented in this paper addresses the gap in using ML methods for estimating building energy loads through a comprehensive study of common ML models fitting over energy simulation data. As became evident in the reviewed literature, despite the wide usage of MLs in this field, a conclusion on selecting the right model for the energy prediction was not possible. The main reason is that most of the research works have focused on the first eminent part of statistical modelling which is features selection. This paper discussed the importance of ML model optimisation in providing a fair comparison of different methods in term of accuracy, the simplicity of tuning and training and response times of model. This study optimised the hyper-parameters of each model for both heating and cooling loads to obtain the best precision. It was also indicated that when there are two energy indices as cooling and heating loads to be estimated by model, it is desired to optimise and train separate machines. To that end, the role of ML model in recognising most impacting factor in prediction of building loads. The other key outcome of this research is a set of recommendations for the quick selection of ML model based on the data and usage.

The results indicated that the standard and advanced GBRTs provide the most accurate predictions, considering the RMSE value. However, when the data was simple (in term of input variables and size), SVM was proven to be the best choice because of simplicity and the speed of calculations. The results also ascertained that for complex data sets, multi-layer NNs are more appropriate when there is a massive demand for ever-more energy simulations. In this case, NN was proven to be capable of estimating incredibly faster than other MLs methods. It should be noted that NN is complicated, and requires an expert to particularly tune it for each studied case; otherwise, NNs could fail quickly.

Comparison of tuned models with previous studies highlighted the importance of determining the hyper-parameters for each data set, and the fact that this can become more crucial by increasing the size and intricacy of the examinations set. By fitting individual models for heating and cooling loads, it was shown that one assorted set of model parameters could not accurately estimate both values. Therefore, unlike previous studies, it is recommended by this study to train models for each

energy load independently. The other approach would be the implementation of a specific sorting algorithm to find balanced values. As results signified, it is suggested to attain a higher accuracy feeding the machines with more number of instances is essential. It might not be a solution for measured historical data; however further simulation using various values of inputs could be aggregated during the design stage prior to optimising the building. Another identified critical factor was that the features must be thoroughly selected/created for representing building characteristics and needs should be appropriately investigated before developing models.

The findings of this study concurred with the seminal literature by demonstrating the fact that MLs techniques are overtly superior over the conventional statistical and engineering methods in building energy calculation. This study also revealed the further power of those ML methods and newly developed ones when they thoroughly optimised. There are several ready to use software packages (e.g. Matlab) providing various ML models with few parameters to modify. Nevertheless, it is advisable to use simpler models like SVM or RF with an advanced programming language, such as Python and R.

Finally, the most important features are recognised using sensitivity analysis methods, and the investigation of the model with reduced dimension revealed that even though the computational cost of building model is reduced, the performance didn't alter. This analysis demonstrated the capability of MLs in eliminating inessential input parameters, while most statistical methods are susceptible to these type of features.

The methods discussed in this work proved the efficiency of ML models in predicting building energy loads as well as performance. The fast and accurate calculation of those values paves pathways for more informed and productive design decisions for built environments. Furthermore, along with the optimisation algorithms, ML seems as a promising solution for efficacious retrofit planning of complex buildings, where engineers are not capable of massive calculations.

Dataset size:
Ecotect dataset: **768 samples**
EnergyPlus dataset: large simulated dataset with multiple building scenarios

Overlap:
Uses ensemble models such as GBRT and RF (same boosting family as LightGBM)

Relevance:
Shows boosting-based models achieve high accuracy in energy prediction

Paper	Method	Dataset Size	Metric	Strength	Weakness	Overlap	Relevance
ASHRAE Great Energy Predictor III	LightGBM, Ensemble Models	1,448 buildings, 20M+ samples	RMSLE	Large real-world dataset, high accuracy	Focused on tabular data only	Gradient Boosting + Feature Engineering	Shows LightGBM works best on real-world data
GBM Lessons Learned (ACM)	LightGBM, GBM, Ensemble	Not explicitly mentioned (ASHRAE dataset)	RMSLE (from competition)	Real-world insights, best practices	Based on one dataset only	Gradient Boosting + Preprocessing	Justifies LightGBM over deep learning
Retrofit Energy Prediction (ASHRAE)	LightGBM, RF, Linear Regression	Not explicitly mentioned	RMSLE	Good performance on real data	Depends on preprocessing quality	LightGBM + Feature Engineering	Shows boosting works in practical scenarios
Heating & Cooling Prediction (LightGBM)	LightGBM + Optimisation	Not explicitly mentioned (simulated data)	RMSE, MAE, R^2	High accuracy after tuning	Uses simulated data	LightGBM + Feature Selection	Shows LightGBM improves prediction accuracy
Springer Paper (LightGBM vs RF vs XGBoost)	LightGBM, RF, XGBoost	Multi-year data (2009–2011)	MAE, RMSE, MSE, R^2	Uses optimisation techniques	Limited dataset scope	Ensemble Models (LightGBM)	Shows LightGBM performs strongly
LightGBM vs XGBoost	LightGBM, XGBoost	Not explicitly mentioned	Accuracy comparison	Direct model comparison	Limited dataset generalisation	Gradient Boosting Models	Shows LightGBM is efficient and competitive

Deep Learning vs ML Paper	DNN, ANN, GB, RF, SVM	~5000 buildings, 285,000 data points	RMSE, MAE, MSE, R^2	Large dataset, many models compared	DL needs more computation	Includes Gradient Boosting	Shows ML can compete with DL
Tuning ML Models (EnergyPlus + Ecotect)	ANN, SVM, RF, GBRT	768 samples + large simulated dataset	RMSE, MAE, R^2	Detailed comparison of models	Simulated data	Ensemble (GBRT similar to LightGBM)	Shows boosting models achieve best accuracy